

Meeting Minutes 5

Project Name: Financial and Operational Analysis of U.S. Hospitals Using Hospital Provider Cost Report Data (2011–2022)

Meeting Minutes 5A (In-Class – Project Interview & Baseline Model Review)
(Feature Importance Validation & External Dataset Justification)

Date and Time

3 Feb 2026, 9:00 AM – 10:00 AM

Location

St. Clair College, One Riverside Drive, Room 1004

Attendees

- Professor Manjari Maheshwari
- Manjunath Muthineni (0878795)
- Rajesh Thota (0873782)
- Neha (0873038)
- Abhishekh Choudhary (0873460)
- Krishna Chaitanya Venuturumilli (0874277)

Absent

None

Agenda

- Present updated baseline ML results
- Review feature importance rankings
- Justify inclusion of Census and BLS datasets
- Discuss next modeling steps

Discussion Points

- During the in-class project interview, **Manjunath** presented updated Random Forest baseline models predicting hospital Profit Margin and Charity Care Ratio (CCR) using the merged hospital dataset.
- Feature importance rankings were reviewed in detail by **Rajesh** to identify the strongest predictive variables.
- BLS healthcare wage variables (physician, RN, practitioner, and support wages) appeared among the top predictors across both models.
- **Neha** highlighted that Census-based socioeconomic indicators such as median income and population also ranked prominently, indicating community-level influence on hospital financial outcomes.
- **Abhishekh** emphasized that the appearance of these external variables empirically validated the integration of Census and BLS datasets into the modeling framework.

- Professor Maheshwari acknowledged that Census and BLS variables were substantively relevant to the research objective.
- It was clarified by **Krishna** that feature importance reflects predictive contribution within the Random Forest model rather than causal inference.

Decisions Made

- Census and BLS datasets formally validated.
- Proceed to refine preprocessing and expand model comparison.

Action Items

- Improve categorical variable handling.
- Implement multi-model testing.

Next Meeting Date and Time

6 Feb 2026 (Group meeting – preprocessing refinement)

Minutes Prepared By

Manjunath Muthineni

Meeting Minutes 5B (Group Meeting – Categorical Encoding Strategy)
(Preprocessing Refinement for Advanced Modeling)

Date and Time

6 Feb 2026, 7:00 PM – 8:30 PM

Location

Online (Microsoft Teams)

Attendees

- Manjunath Muthineni (0878795)
- Rajesh Thota (0873782)
- Neha (0873038)
- Abhishekh Choudhary (0873460)
- Krishna Chaitanya Venuturumilli (0874277)

Absent

None

Agenda

- Identify categorical variables
- Select encoding strategy
- Prepare dataset for advanced modeling

Discussion Points

- **Manjunath** identified categorical variables including provider_type, ccn_facility_type, type_of_control, rural_urban, state_code, fips, and medicare_cbsa_number from the merged dataset.
- Risks of ordinal bias from label encoding were discussed by **Rajesh**, particularly for nominal hospital classification variables without natural ordering.
- **Neha** recommended selecting One-Hot Encoding for nominal categories to preserve category independence.
- **Abhishekh** confirmed maintaining an 80/20 train-test split and random_state = 42 for reproducibility across experiments.
- **Krishna** agreed on maintaining consistent preprocessing steps across all advanced models.

Decisions Made

- Apply One-Hot Encoding using category_encoders.
- Maintain consistent preprocessing across all models.

Action Items

- Retrain models after encoding.

Next Meeting Date and Time

10 Feb 2026 (Group meeting – multi-model testing)

Minutes Prepared By

Rajesh Thota

Meeting Minutes 6

*Meeting Minutes 6A (Group Meeting – Multi-Model Advanced Regression Testing)
(Preliminary Model Comparison)*

Date and Time

10 Feb 2026, 7:00 PM – 9:00 PM

Location

Online (Microsoft Teams)

Attendees

- Manjunath Muthineni (0878795)
- Rajesh Thota (0873782)
- Neha (0873038)
- Abhishekh Choudhary (0873460)
- Krishna Chaitanya Venuturumilli (0874277)

Absent

None

Agenda

- Implement multiple regression models
- Compare predictive performance

Discussion Points

- **Manjunath** implemented multiple regression models for Profit Margin and CCR, including Random Forest, Gradient Boosting, Extra Trees, Ridge, Lasso, XGBoost, and LightGBM.
- Model performance was evaluated using test R^2 by **Rajesh Thota** to compare predictive accuracy across linear, regularized, and ensemble methods.
- **Neha** observed that ensemble models significantly outperformed linear models, indicating nonlinear relationships and interactions within hospital financial data.
- **Abhishekh** noted that current results were based on default hyperparameter settings and therefore represent preliminary model performance.

Decisions Made

- Retain ensemble models as leading candidates.
- Proceed to feature reduction for improved robustness.

Action Items

- Prepare comparative performance table.
- Implement Sequential Feature Selection.

Next Meeting Date and Time

13 Feb 2026 (Group meeting – feature selection & validation)

Minutes Prepared By

Krishna Chaitanya Venuturumilli

Meeting Minutes 7

*Meeting Minutes 7A (Group Meeting – Sequential Feature Selection & Cross-Validation)
(Dimensionality Reduction & Model Stabilization)*

Date and Time

13 Feb 2026, 7:00 PM – 8:45 PM

Location

Online (Microsoft Teams)

Attendees

- Manjunath Muthineni (0878795)
- Rajesh Thota (0873782)
- Neha (0873038)
- Abhishekh Choudhary (0873460)
- Krishna Chaitanya Venuturumilli (0874277)

Absent

None

Agenda

- Reduce numerical feature space
- Improve stability through cross-validation

Discussion Points

- Sequential Feature Selection reduced numerical predictors to 15 features per target, implemented by **Manjunath** to remove redundant predictors.
- **Rajesh** applied 3-fold cross-validation during selection and 5-fold CV for evaluation to ensure robustness.
- **Neha** observed that feature reduction improved model stability and interpretability of financial predictors.
- **Abhishekh** confirmed that feature selection was performed using training data only, ensuring no data leakage occurred.

Decisions Made

- Freeze selected feature subsets pending hyperparameter tuning.

Action Items

- Document selected features.
- Prepare in-class review summary.

Next Meeting Date and Time

17 Feb 2026 (In-class – advanced results review)

Minutes Prepared By

Neha

Meeting Minutes 8

Meeting Minutes 8A (In-Class – Advanced Model Review & Interim Report Guidance)

Date and Time

17 Feb 2026, 9:00 AM – 10:00 AM

Location

St. Clair College, One Riverside Drive, Room 1004

Attendees

- Professor Manjari Maheshwari
- Manjunath Muthineni (0878795)
- Rajesh Thota (0873782)
- Neha (0873038)
- Abhishekh Choudhary (0873460)
- Krishna Chaitanya Venuturumilli (0874277)

Absent

None

Agenda

- Present advanced model comparison results
- Discuss interim ML report structure
- Clarify final model selection process

Discussion Points

- Preliminary test R^2 results were presented by **Manjunath**; Random Forest currently demonstrates the highest performance for both targets.
- **Rajesh** noted that model comparisons remain preliminary because all models were trained with default hyperparameters.
- Professor Maheshwari emphasized that final model selection should not be confirmed without hyperparameter tuning.
- Academic articles were provided to guide interim report structure.

Decisions Made

- Random Forest identified as leading candidate (preliminary).
- Hyperparameter tuning required before final selection.

Action Items

- Conduct hyperparameter tuning experiments.
- Update interim report to reflect preliminary results.

Next Meeting Date and Time

24 Feb 2026 (Scheduled – tuning methodology discussion)

Minutes Prepared By

Abhishekh Choudhary

Meeting Minutes 9

Meeting Minutes 9A (Scheduled In-Class – Hyperparameter Tuning Plan & Model Selection Criteria)

Date and Time

24 Feb 2026, 9:00 AM – 10:00 AM

Location

St. Clair College, One Riverside Drive, Room 1004

Status

Scheduled / Upcoming

Proposed Agenda

- Review preliminary model comparison
- Discuss hyperparameter tuning methodology
- Define final model selection criteria
- Review interim report progress

Planned Discussion Points

- **Manjunath** will review preliminary model comparison results.
- **Rajesh** will discuss hyperparameter tuning methodology.
- **Neha** will identify key parameters for tuning (`n_estimators`, `max_depth`, `learning_rate`, `min_samples_split`, `subsample`, `regularization`).
- **Abhishekh** will ensure fair comparison across tuned ensemble models.
- **Krishna Chaitanya** will define final model selection criteria based on cross-validated R^2 , stability, and generalization.
- Current results will be clearly labeled as preliminary in the interim report.

Prepared By

Krishna Chaithanya Venuturumilli